

Mathematical Physics I (Fall 2026): Homework #2

Due Oct. 2, 2026 (Fri, 23:00pm)

[0.5 pt each, total 5 pts, turn in as a single pdf file to eTL]

- By turning in your homework, you acknowledge that you have not received any unpermitted aid, nor have compromised your academic integrity during its preparation. (Remember the SNU College of Natural Sciences Honor Code, and the SNU AI Guideline!)
- Textbook problem numbering convention (Boas, 3rd ed.): [Section].[Problem No.]. Many problems in this assignment are from the list of suggested problems announced in the class.
- Exhibit all intermediate steps to receive full credits. Only handwritten answers are accepted except for numerical problems or verifications — for which you print out and turn in not just the end results (e.g., plots) but also the source codes in your favorite tools such as MATLAB or MATHEMATICA.

1.-8. Boas Chapter 3, Problems 5.37, 5.38, 6.16, 7.27, 8.15, 9.17, 11.44, 12.16

(Note: For Problem 5.37, review the Examples in Boas Chapter 3, Section 5 in case you have not done it yet. Find the most relevant example and use the method depicted in there. Solve the problem in two different ways, following the two suggestions given. For the second method, start with the two points: $P(3, -2, 1)$ on the first line and $Q(5, -2, 0)$ on the second line. For Problem 5.38, find the acute angle between the lines. For Problem 6.16, find the inverse with two different methods — using Eq.(6.13) of Boas Chapter 3 and using the *Gauss-Jordan matrix inversion procedure*. Compare your results with the one found with a computer.)

9. The polarization of electromagnetic waves and the properties of optical filters can be described using vectors and matrices.

(a) A beam of linearly polarized light is described by the two-component vector

$$\mathbf{E} = \begin{pmatrix} E_x \\ E_y \end{pmatrix}, \quad (1)$$

where E_x and E_y are the amplitudes of the electric field along two perpendicular directions, x and y . Meanwhile, an ideal linear polarizer whose transmission axis makes an angle θ with the x -axis is represented by the matrix

$$P(\theta) = \begin{pmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{pmatrix}.$$

Show that $P(\theta)^2 = P(\theta)$, and explain why this property is expected for an *ideal* polarizer.

(b) Find the eigenvalues and normalized eigenvectors of $P(\theta)$, and interpret the two eigenvectors physically.

(c) Consider two ideal polarizers. The first has its transmission axis along the x -axis, while the second is rotated by an angle θ relative to the first. Then the electric field after passing through the two polarizers can be written as $\mathbf{E}_{\text{out}} = P(\theta)P(0)\mathbf{E}_{\text{in}}$. Given the incident light in Eq.(1), find $\mathbf{E}_{\text{out}}$ explicitly. Show that the intensity of the transmitted light is $I_{\text{out}} = I_0 \cos^2 \theta$, where I_0 is the intensity after the first polarizer.

(d) Now consider three ideal polarizers. The first and last polarizers are crossed, while the middle polarizer is oriented at an angle θ relative to the x -axis:

$$P(0), \quad P(\theta), \quad P\left(\frac{\pi}{2}\right).$$

Show that although the first and last polarizers are crossed and would completely block the light if placed directly together, a nonzero intensity can emerge when the intermediate polarizer is inserted. Find the transmitted intensity as a function of θ and determine the angle at which the transmitted intensity is maximized.

(Note: For (c) and (d), you may want to briefly review the freshman physics textbooks such as Halliday & Resnick Chapter 33.4.)

10. In the class we discussed the Gram-Schmidt orthonormalization process for a linear vector space and for a general vector space.

(a) In Example 6 of Boas Chapter 3, Section 14 — which we briefly discussed in class but left for you to work through — with an inner product defined as

$$\langle f|g \rangle = \int_{-1}^1 f^*(x)g(x)dx,$$

one can start with the functions $f_i = x^i$ ($i = 0, 1, 2, 3$) and construct a set of orthonormal polynomials P_i that satisfy the orthonormality condition on the interval $-1 \leq x \leq 1$,

$$\int_{-1}^1 P_m(x)P_n(x)dx = \delta_{mn}$$

(see also Eq.(8.4) of Chapter 12, but note a different normalization factor). We later identify this set of functions as the *Legendre polynomials*. Starting from Eq.(14.10), follow the procedure step by step and find for yourself the first four members of P_i .

(b) Consider a different set of functions defined with a similar inner product as in (a), but over a shifted interval $0 \leq x \leq 1$. The orthonormality condition is now given by

$$\int_0^1 P'_m(x)P'_n(x)dx = \frac{1}{2m+1} \delta_{mn}.$$

Find the first three members in the set of orthonormal polynomials P'_i (the sign ' does not mean differentiation). We refer to this set of functions as the *shifted Legendre polynomials*.

(c) Now, with a new inner product defined as

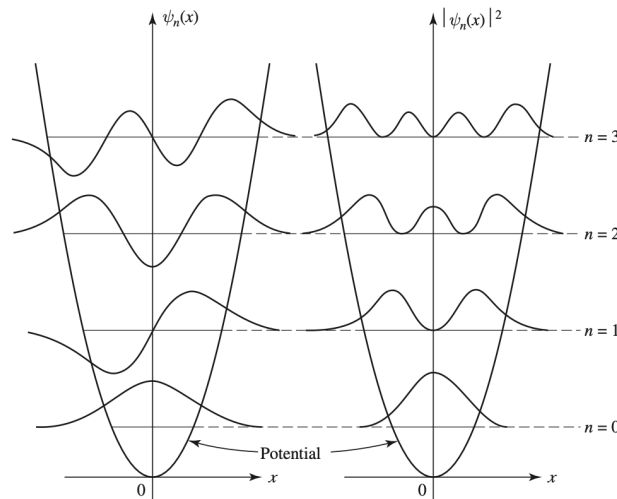
$$\langle f|g \rangle = \int_{-\infty}^{\infty} f^*(x)g(x)e^{-x^2}dx, \quad \text{where } e^{-x^2} \text{ is the "weighting" function,}$$

find the first three members in the set of orthonormal polynomials H_i that satisfy the orthonormality condition on the interval $-\infty \leq x \leq \infty$,

$$\int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx = \delta_{mn} \sqrt{\pi} 2^m m!$$

(see also Eq.(22.15) of Chapter 12). We refer to this set of functions as the *Hermite polynomials*.

(d) Discuss briefly where the (associated) Legendre polynomials and the Hermite polynomials appear in quantum mechanics or in other physics research.



[Adapted from Griffiths & Schroeter, Chapter 2.3]

(Note: For (c), you may simply use the value of the so-called Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$ without proof, although it can be easily found using the techniques in Boas Chapter 5, Section 4, or in Problem 9.4 of Chapter 11. The Gaussian integral appears frequently also in elementary calculus textbooks such as Stewart (Chapter 15.3) or 김홍중 (Chapter 14.4). However, all other integrals such as $\int_{-\infty}^{\infty} x^2 e^{-x^2} dx$ must be evaluated explicitly, for example, by using integration by parts or the technique in Problem 12.16 of Boas Chapter 4. For (d), you may want to briefly review the textbooks in quantum mechanics such as Griffiths & Schroeter. Once again, you are simply asked to come up with 2-3 sentences about how these polynomials are used, not the detailed physical or mathematical discussions. *Astronomy majors* are highly encouraged to look for the use cases of Legendre polynomials in astrophysics and cosmology.)